

TurboADMM-NL: Nested Parametric Continuation for Nonlinear Heterogeneous Multi-Agent Trajectory Optimization

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Abstract—Nonlinear multi-agent trajectory optimization repeatedly solves coupled quadratic programs whose temporal dynamics, collision geometry, and active constraints evolve gradually. General-purpose centralized solvers do not exploit agent separability, while conventional distributed methods repeatedly discard useful state at nonlinear relinearization boundaries. This letter extends the linear-QP solver TurboADMM to TurboADMM-NL, a shared-memory sequential convex programming method for heterogeneous robot teams. The method combines per-agent alternating-direction decomposition with four levels of structure reuse: affine Riccati initialization, vector hotstarts within each ADMM solve, matrix and working-set continuation across nonlinear linearizations, and geometry-aware transport of pairwise consensus and dual state. A merit line search, trust-region restoration, elastic feasibility recovery, and an independent continuous validator prevent continuation from weakening nonlinear safety. The same problem may contain unicycles, bicycles, and reduced-order three-dimensional quadcopters and convex polygonal obstacles. Frozen deterministic and Monte Carlo protocols compare against centralized qpOASES and OSQP and isolate each continuation layer. A disjoint CSDO-compatible evaluation further tests a complementary property: repairing locally conflicted trajectories from a common PBS/Hybrid-A* front end while retaining negotiable inter-agent boundaries. On the untouched 720-case split, TurboADMM-NL validates every case, stays within 4.990% of OSQP, and at $N = 20$ is $2.530\times$ faster with $17.8\times$ lower peak memory. It repairs 32/32 CSDO-compatible cases, versus 15/32 for original CSDO.

Index Terms—Multi-robot systems, trajectory optimization, distributed optimization, sequential convex programming, ADMM.

I. INTRODUCTION

Multi-robot planning couples long-horizon nonlinear dynamics to pairwise collision avoidance. A centralized transcription exposes all decision variables and pair constraints to one optimizer. This is robust at small scale, but the joint QP grows with the team and ignores the per-agent temporal structure. Distributed optimization exposes parallelism, yet its many local QP solves can erase that benefit when each solve is initialized independently. The challenge becomes sharper inside sequential convex programming (SCP): an ADMM loop solves one convexification, after which changed Jacobians and separating planes conventionally trigger another cold optimization.

TurboADMM [1] addressed a single sequence of linearly constrained multi-agent QPs. It combined agent-level ADMM, Riccati-based primal-dual initialization, and qpOASES active-set hotstarts. That architecture reduced local QP work substantially, but it did not address nonlinear dynamics, heterogeneous state dimensions, static obstacles,

changing collision geometry, nonlinear globalization, or recovery from an infeasible reference. Merely placing the previous solver inside an SCP loop would discard the active-set and consensus structure exactly where the nonlinear solver spends most of its work.

This letter introduces TurboADMM-NL, a nonlinear extension that treats the nested SCP-ADMM iteration as a two-dimensional parametric sequence. At fixed linearization, only consensus-dependent QP vectors change. Across SCP iterations, dynamics matrices and collision normals change gradually. We reuse the appropriate state at both boundaries and reset it only when a measurable compatibility test fails. The method is algorithmically distributed but runs on one shared-memory machine; it is not a network-distributed control scheme.

The contributions are:

- A structure-preserving SCP decomposition for teams with mixed nonlinear models, agent-specific dimensions and footprints, convex polygonal obstacles, and inter-knot collision checking.
- Nested parametric continuation that combines Riccati initialization, inner-ADMM vector hotstarts, cross-SCP matrix/working-set continuation, and geometry-aware transport of pairwise auxiliary and scaled-dual state.
- A strict nonlinear recovery mechanism combining exact rollout, merit globalization, cold reliable QP reconstruction, elastic slacks, trust-region contraction, and final fixed-penalty consensus polishing.
- A frozen evaluation protocol with six hand-designed cases, a 24-cell Monte Carlo scaling matrix, centralized qpOASES and OSQP baselines, continuation ablations, and a disjoint paired CSDO recovery study.

The novelty is not ADMM, Riccati recursion, active-set hotstarting, or SCP in isolation. It is their continuation-aware composition across both levels of a nonlinear multi-agent solver, together with evidence that the added reuse reduces work without relaxing feasibility or objective-quality gates.

II. RELATED WORK

Sparse first-order QP solvers such as OSQP [2] are robust general baselines, and parametric active-set solvers such as qpOASES [3] can exploit similarity between consecutive QPs. Optimal-control solvers exploit block temporal structure, but a monolithic multi-agent transcription still exposes growing coupling. Our focus is orthogonal: retain separate per-agent optimal-control QPs and move the coupling into low-dimensional consensus projections.

ADMM provides a standard mechanism for decomposing convex programs with coupling constraints [4]. Multi-agent variants range from physically distributed MPC and DDP architectures [5], [6] to centralized shared-memory decomposition. The latter is the relevant comparison here: communication latency is absent, so performance depends on local QP work, parallel scheduling, and the number of consensus rounds. TurboADMM [1] showed that Riccati initialization and parametric active-set hotstarts are complementary for linear multi-agent QPs. TurboADMM-NL extends that mechanism across nonlinear relinearizations rather than claiming a new ADMM splitting.

Recent decentralized planners make different coupling choices. MADER and Robust MADER [7], [8] optimize against communicated trajectories, with the latter explicitly protecting asynchronous execution from communication delay. EGO-Swarm [9] couples per-agent nonlinear optimization to lightweight topological recovery. Spatiotemporal-occupancy methods [10] embed shared trajectories into local maps and construct independent safe corridors, while MA-DV²F [11] generates dynamic velocity fields independently for nonholonomic vehicles. These methods prioritize reactive or onboard decentralized navigation. In contrast, TurboADMM-NL is a batch optimizer on one shared-memory machine: it retains negotiable pair constraints during every SCP step, but does not claim communication-delay robustness or feedback-control rates.

Sequential convex optimization is widely used to obtain locally optimal collision-free trajectories [12], [13]. Its performance and robustness depend on the reference homotopy, trust-region policy, and recovery from temporarily infeasible convexifications. We preserve a nonlinear merit test and independently validate every returned trajectory; a feasible SCP-limit iterate is reported separately from strict convergence.

Search-then-refine pipelines also smooth conflict-free discrete schedules for large quadrotor teams [14]. CSDO [15] combines centralized priority-based search (PBS) and spatiotemporal Hybrid-A* with decentralized QP refinement. Its separating planes and safe corridors are constants derived from the initial guess, so each vehicle QP is independent. This makes refinement extremely fast when the searched homotopy and assignment of avoidance responsibility are already adequate. Our CSDO experiment keeps the same search front end but asks a different question: can a coupled optimizer repair a locally conflicted root by negotiating pairwise motion rather than accepting fixed mutual boundaries? Runtime is therefore secondary in that comparison and is disclosed as a non-advantage.

III. NONLINEAR MULTI-AGENT PROBLEM

Consider N agents over horizon H . Agent i has state $x_{i,k} \in \mathbb{R}^{n_i}$, input $u_{i,k} \in \mathbb{R}^{m_i}$, discrete dynamics f_i , and position output h_i . An embedding E_i maps a two- or three-dimensional position into the common workspace $\mathbb{R}^d$, where

TABLE I
IMPLEMENTED HETEROGENEOUS AGENT MODELS.

Model	n_i	m_i	Position
Unicycle	4	2	(p_x, p_y)
Bicycle	5	2	(p_x, p_y)
Reduced quadcopter	9	4	(p_x, p_y, p_z)

$d = \max_i \dim h_i$. The dynamics and position maps are

$$x_{i,k+1} = f_i(x_{i,k}, u_{i,k}), \quad (1)$$

$$p_{i,k} = E_i h_i(x_{i,k}). \quad (2)$$

Each agent has radius r_i ; ground footprints are discs and aerial footprints are spheres. Polygonal obstacles are treated as vertical prisms when $d = 3$.

With local stage and terminal costs ℓ_i and ϕ_i , the joint problem is

$$\min_{x,u} \sum_{i=1}^N \left[\phi_i(x_{i,H}) + \sum_{k=0}^{H-1} \ell_i(x_{i,k}, u_{i,k}) \right] \quad (3a)$$

$$\text{s.t. } x_{i,k+1} = f_i(x_{i,k}, u_{i,k}), \quad (3b)$$

$$(x_{i,k}, u_{i,k}) \in \mathcal{X}_{i,k} \times \mathcal{U}_{i,k}, \quad (3c)$$

$$\|p_{i,k} - p_{j,k}\|_2 \geq r_i + r_j, \quad i < j, \quad (3d)$$

$$\text{dist}(p_{i,k}, \mathcal{O}_q) \geq r_i, \quad (3e)$$

together with masked terminal equalities or terminal tolerances. Collision constraints are enforced both at knots and interpolated substeps. Problem (3) is nonconvex because of the dynamics and reverse-convex separation constraints.

The implementation supports the model set in Table I. A single problem may mix all three. The quadcopter is deliberately reduced order; a full rigid-body torque model is outside the present claim.

IV. TURBOADMM-NL

A. Structure-Preserving Convexification

At SCP iteration s , let $(\bar{x}^s, \bar{u}^s)$ be the nominal trajectory and use increments $(\delta x, \delta u)$. Linearized dynamics are

$$\delta x_{i,k+1} = A_{i,k}^s \delta x_{i,k} + B_{i,k}^s \delta u_{i,k} + d_{i,k}^s, \quad (4)$$

where $A_{i,k}^s$ and $B_{i,k}^s$ are analytic Jacobians and $d_{i,k}^s$ is the affine defect. For pair (i, j) , define nominal displacement $\bar{q}_{ij,k}^s = \bar{p}_{i,k}^s - \bar{p}_{j,k}^s$ and unit normal $n_{ij,k}^s = \bar{q}_{ij,k}^s / \|\bar{q}_{ij,k}^s\|_2$. The supporting half-space is

$$(n_{ij,k}^s)^T (J_{i,k}^s \delta x_{i,k} - J_{j,k}^s \delta x_{j,k}) \geq r_i + r_j - (n_{ij,k}^s)^T \bar{q}_{ij,k}^s. \quad (5)$$

Convex polygon constraints use supporting planes on clearance-inflated boundaries. A proximity margin screens inactive pairs and obstacles in the constant-density experiments; the nonlinear merit and independent validator still evaluate every pair and obstacle.

B. Agent-Level ADMM

Collect agent i 's increments in w_i . The convexified objective and local constraints give

$$\min_{w_i \in \mathcal{W}_i^s} \frac{1}{2} w_i^T H_i^s w_i + (g_i^s)^T w_i, \quad (6)$$

where $\mathcal{W}_i^s$ contains (4), bounds, terminal rows, trust-region rows, and static-obstacle half-spaces. For each active pair-stage incidence e , a selector $P_{i,e}^s$ exposes the local position used by (5). Introducing auxiliary positions z_e and scaled duals $y_{i,e}$ yields the local ADMM update

$$w_i^{r+1} = \arg \min_{w_i \in \mathcal{W}_i^s} \frac{1}{2} w_i^T H_i^s w_i + (g_i^s)^T w_i + \frac{\rho}{2} \sum_{e \in \mathcal{E}_i^s} \|P_{i,e}^s w_i - z_{i,e}^r + y_{i,e}^r\|_2^2. \quad (7)$$

All local QPs are independent and are solved in parallel. The coordinator projects each two-agent auxiliary block onto its collision half-space and then updates the scaled duals. Thus no centralized coupling QP is solved.

C. Nested Parametric Continuation

The architecture reuses structure at four distinct boundaries.

- 1) *Riccati initialization.* At the first local solve, an affine, stage-varying LQR recursion produces a dynamics-feasible primal trajectory and equality multipliers in $O(H)$ stage operations. Inequality multipliers begin at zero.
- 2) *Inner vector hotstart.* Within one SCP linearization, the Hessian and constraint matrix of (7) are fixed while the consensus-dependent gradient and bounds vary. qpOASES reuses the previous primal-dual point and active set.
- 3) *Cross-SCP matrix hotstart.* When A^s , B^s , obstacle rows, or collision rows change, qpOASES receives the new matrices while retaining the compatible working set and local solution.
- 4) *Geometry-aware consensus transport.* A pair-stage state is transported only if $(n_{ij,k}^{s-1})^T n_{ij,k}^s \geq \gamma$. Its auxiliary positions are projected onto the new half-space and only the dual component aligned with $n_{ij,k}^s$ is retained. Incompatible states are reset from the nominal trajectory.

These mechanisms define four controlled ablations: *cold*, *inner*, *QP continuation*, and *full*. All variants share the same model, linearization, cost, initialization, penalty, trust region, and tolerances. Continuation changes only the initialization of the local active-set and ADMM iterations. Conditioned on convergence of a convexified subproblem, every variant therefore satisfies the same primal feasibility and ADMM fixed-point conditions; transported pair state neither changes the objective nor removes a constraint. Projection of compatible auxiliaries and reset of incompatible state preserve this invariant across relinearizations. Nonlinear feasibility is then enforced separately by globalization and the independent validator.

D. Globalization and Feasibility Recovery

A nonlinear merit line search accepts an SCP candidate only when the true rollout improves the frozen merit. Full-state references are projected through the exact nonlinear dynamics whenever this restores dynamics without violating physical safety. If a convex subproblem fails, the solver rebuilds it cold with reliable active-set settings. A second fallback introduces nonnegative elastic slacks on temporarily infeasible obstacle, corridor, and terminal rows, penalized by a fixed exact-penalty weight. Rejected steps roll back to the last accepted nonlinear iterate and contract the control trust region.

After the ordinary convergence predicate first holds, a fixed- ρ phase polishes consensus with zero relative residual tolerance. Restoration slacks must vanish, the QP direction must meet the unchanged SCP step tolerance, and the nonlinear trajectory must satisfy pair, obstacle, dynamics, and terminal tests before strict convergence is declared.

E. Structural Scaling

Let $q_i = O(H(n_i + m_i))$ be one local QP dimension, Δ the maximum active interaction degree, and P the number of worker threads. One ADMM round costs approximately

$$O\left(\frac{1}{P} \sum_i C_{QP}(q_i) + H|\mathcal{E}^s|d\right), \quad (8)$$

whereas a centralized solver processes dimension $\sum_i q_i$ and all coupling rows simultaneously. TurboADMM-NL does not claim a universal asymptotic bound for active-set work. Its intended regime is spatially local interaction, where Δ and local QP dimensions remain bounded while the centralized problem continues to grow. The experiments report active degree, QP dimensions, QP solves, working-set recalculations, solver time, process wall time, and peak memory rather than inferring scalability from runtime alone.

V. EXPERIMENTAL PROTOCOL

All publication runs use the same WSL Release binary and independent nonlinear validator. Collision and obstacle clearance are sampled at 20 substeps per interval. A run succeeds only if solver execution finishes under the frozen cap and the validator accepts pair clearance, obstacle clearance, nonlinear dynamics, and terminal constraints. Strict SCP convergence is reported separately and is required for every Turbo deterministic and development-gate case. Run manifests hash the executable, runner, scenario inventory, method order, and OpenMP policy.

A. Comparisons and Ablations

The internal comparisons are centralized qpOASES and official OSQP applied to the identical SCP QP, plus the cold, inner, QP-continuation, and full distributed variants. Deterministic timings use ten fresh processes with interleaved method order. Monte Carlo uses paired seeds and reports medians, quartiles, Wilson success intervals, exact paired sign tests for crossover, and objective gaps on jointly valid outputs. All six variants run on development; the untouched

final split repeats only full, centralized OSQP, and centralized qpOASES because the ablation is frozen on 240 development cases.

B. Scenario Sets

The six headline deterministic cases progress from a two-unicycle open crossing to a single blocker, a four-agent doorway, a heterogeneous doorway, an eight-agent warehouse, and an eight-agent maze. A heterogeneous open case remains a regression-only overhead control and is not used as headline performance evidence.

The submission-primary Monte Carlo grid contains $N \in \{4, 8, 12, 20\}$, obstacle counts $M \in \{0, N, 2N\}$, homogeneous unicycles and balanced heterogeneous teams, and ten development or thirty untouched final seeds per cell. Constant-density local exchanges provide an analytic feasibility witness and are designed to keep active interaction degree bounded. This gives 240 development and 720 final scenarios.

For CSDO, both branches use the untouched, deterministically seeded PBS/Hybrid-A* front end. On search success, CSDO’s pre-DSQP initial guess is passed directly to TurboADMM-NL. A clean-source exporter repeats the same seeded search, verifies schedule equality to 10^{-9} , and reconstructs the static obstacle corridors before either optimization branch is run. Original CSDO then constructs fixed inter-agent separating planes and solves independent vehicle QPs; TurboADMM-NL instead retains negotiable pair coupling. If PBS fails, the exporter preserves a conflicted root when PBS exposes one. If PBS exposes no root, it runs the same Hybrid-A* search independently for each vehicle, producing individually feasible but mutually conflicted paths. Either failure artifact and its reconstructed static corridors become TurboADMM-NL’s input, while original CSDO reports search failure. TurboADMM-NL may widen only the temporal indexing of the reconstructed static corridor under the frozen recovery rule. Every source, executable, schedule, and corridor artifact is hashed. Development uses 24 passing-bay instances; evaluation uses a disjoint 32-case grid varying bay length, arrival offset, and priority order.

VI. RESULTS

A. Deterministic Stress Tests

TurboADMM-NL validates and strictly converges on 60/60 runs. Its maximum absolute objective gap on jointly valid OSQP outputs is 2.319%. The maximum terminal error and dynamics defect are 3.01×10^{-1} and 0; minimum sampled pair and obstacle clearances are 1.22×10^{-7} and -1.43×10^{-3} , accepted under the validator’s fixed 10^{-2} numerical feasibility tolerance.

B. Untouched Final Evaluation

The untouched final split tests the predeclared structural crossover. On 720 paired scenarios, both TurboADMM-NL and centralized OSQP return independently valid, strictly converged trajectories. The median OSQP/TurboADMM-NL process-wall ratios are 0.488, 0.928, 1.627, and 2.530 for

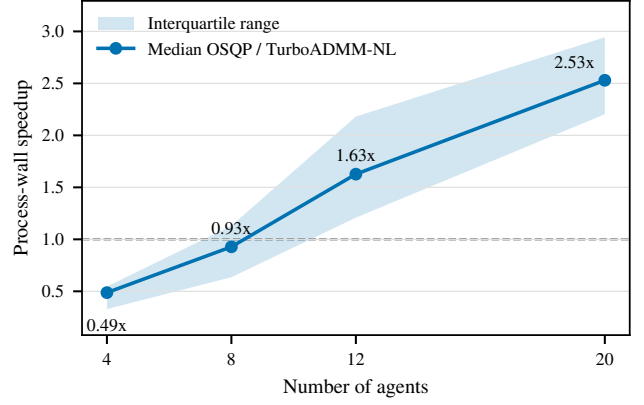


Fig. 1. Final scaling on 180 paired cases per team size. Values above one favor TurboADMM-NL; shading is the interquartile range.

$N = 4, 8, 12, 20$, respectively. The crossover occurs between $N = 8$ and $N = 12$ and is statistically significant at $N = 12$ ($p = 2.15 \times 10^{-20}$) and $N = 20$ ($p = 1.31 \times 10^{-54}$). At $N = 20$, centralized OSQP uses $17.8\times$ the median peak memory. The median, 95th-percentile, and maximum absolute objective gaps are 0.214%, 4.475%, and 4.990%, respectively, below the predeclared 5% gate.

The continuation ablation contains 240 common scenarios for each distributed variant. Matrix/working-set continuation reduces median runtime by 45.6% relative to inner vector hotstarts alone. Adding geometry-aware pair-state transport yields a further 6.3% overall and 12.6% at $N = 20$, while reducing the median number of local QP solves by 38.5%. Together, the full cross-SCP continuation stack reduces median runtime by 46.6%. Every variant validates and strictly converges on all 240 cases.

C. CSDO-Compatible Recovery

Table IV reports every case in the disjoint evaluation grid. TurboADMM-NL validates 32 of 32 cases, versus 15 of 32 for original CSDO. The paired outcomes contain 15 shared successes, 17 TurboADMM-NL-only successes, 0 CSDO-only successes, and 0 joint failures. The exact two-sided McNemar test is $p = 1.53 \times 10^{-5}$. Of 17 cases where CSDO’s PBS search fails, TurboADMM-NL repairs 17 (100.0%), while CSDO repairs none. These comprise 2 retained-root cases and 15 cases initialized from mutually conflicted single-agent Hybrid-A* paths. Every success is checked by the same continuous validator.

The recorded times do not have identical scope and therefore are not used for a runtime comparison. CSDO’s searched separating planes eliminate all optimization coupling, whereas TurboADMM-NL retains and negotiates pair constraints. This experiment isolates the resulting recovery capability, not speed. On shared successes, median path length differs by only 1.5%, so we also make no broad reduced-conservatism claim from this suite.

TABLE II

FROZEN DETERMINISTIC STRESS TESTS. EACH ENTRY IS VALID RUNS/10; MEDIAN WSL PROCESS WALL TIME IN SECONDS AMONG VALID RUNS. PARENTHESES IN THE SCENARIO COLUMN GIVE AGENTS/OBSTACLES. THE EXECUTION CAP IS 300 S; “–” DENOTES NO VALID COMPLETION.

Scenario (N/M)	TurboADMM-NL	OSQP	qpOASES
Easy open (2/0)	10/10; 0.68	10/10; 0.31	10/10; 0.87
Easy blocker (2/1)	10/10; 1.12	10/10; 1.07	10/10; 1.82
Medium doorway (4/2)	10/10; 3.65	10/10; 26.46	10/10; 19.16
Heterog. doorway (4/2)	10/10; 8.94	10/10; 116.48	10/10; 105.54
Warehouse (8/8)	10/10; 54.15	0/10; –	0/10; –
Maze (8/16)	10/10; 67.93	0/10; –	0/10; –

TABLE III

UNTouched FINAL MATRIX. ENTRIES ARE VALID TRAJECTORIES/ 180 AND MEDIAN WSL PROCESS WALL TIME IN SECONDS AMONG VALID RUNS. THE EXECUTION CAP IS 120 S; “–” DENOTES NO VALID COMPLETION.

N	TurboADMM-NL		OSQP		qpOASES	
	Valid	Wall	Valid	Wall	Valid	Wall
4	180/180	1.55	180/180	0.70	180/180	8.92
8	180/180	2.91	180/180	2.02	90/180	46.14
12	180/180	4.01	180/180	5.50	0/180	–
20	180/180	5.19	180/180	12.03	0/180	–

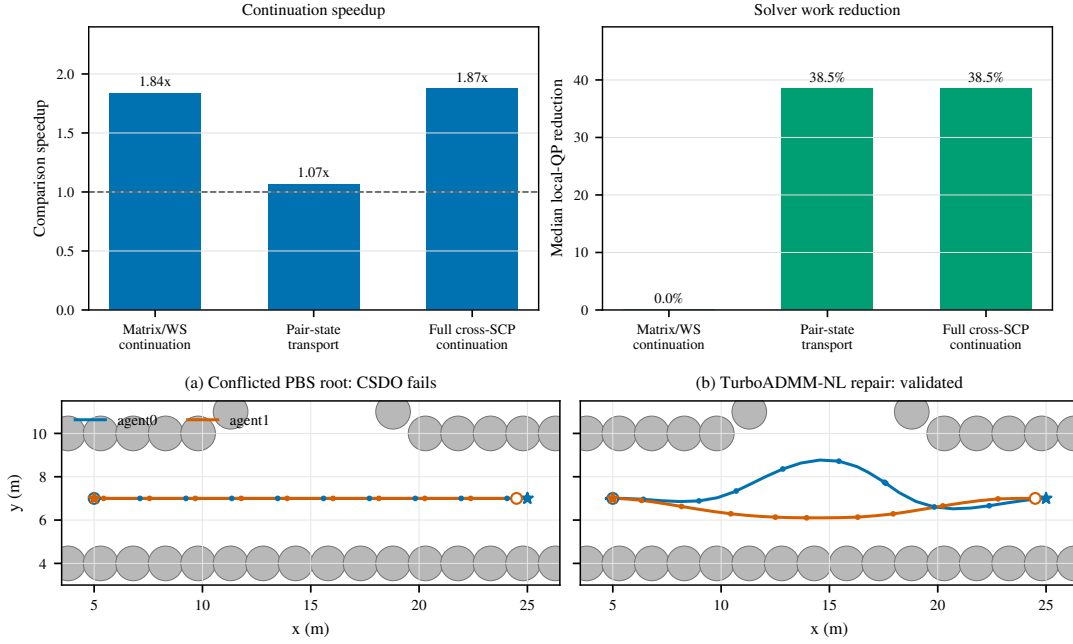


Fig. 2. Top: causal continuation ablation on the development matrix, isolating matrix/working-set reuse, pair-state transport, and their combined cross-SCP effect. Bottom: a representative frozen CSDO-compatible recovery. PBS terminates with mutually conflicted centerline trajectories (left); starting from that exact artifact and its static corridors, TurboADMM-NL negotiates which vehicle enters the bay and returns an independently validated plan (right). Circles are CSDO map obstacles and dots are equal-time samples.

D. Final Protocol Audit

The frozen manifest accounts for all 3×720 scheduled external-method executions as completions or time-out-censored outcomes. All TurboADMM-NL and OSQP rows are independently valid and strictly converged, the objective gate is satisfied, and the generated tables retain every qpOASES outcome. Solver and benchmark parameters were not changed after the final split was inspected.

VII. LIMITATIONS

TurboADMM-NL is a local nonlinear optimizer and does not replace global homotopy search. Its deterministic polygon bypass is suitable for the tested convex obstacles; complicated maps still benefit from a feasible reference or a search front end. The quadcopter model omits full rigid-body angular dynamics. Shared-memory decomposition does not address communication delay or packet loss. Finally, continuation can reduce repeated QP work but does not guarantee fewer SCP or ADMM iterations on every small problem;

TABLE IV
FROZEN PAIRED CSDO EVALUATION (32 CASES).

Metric	CSDO	TurboADMM-NL
Independently valid	15/32	32/32
PBS-failure recovery	0/17	17/17
Method-only successes	0	17
95% Wilson interval	[30.9,63.6]%	[89.3,100.0]%

the claimed advantage is the bounded-degree medium-to-large regime and the ability to negotiate coupling that fixed-corridor refiners remove.

VIII. CONCLUSION

This letter extends TurboADMM from structured linear multi-agent QPs to nonlinear heterogeneous trajectory optimization. The key mechanism is nested parametric continuation: temporal Riccati structure, active-set information, and geometry-aligned consensus state are reused at the iteration boundary where each is valid. Strict globalization and independent validation preserve safety. The frozen experiments are designed to test two complementary claims rather than one universal speed claim: structural crossover against centralized QP solvers and recovery of locally conflicted search trajectories that fixed separating planes cannot negotiate.

REFERENCES

- [1] Y. Chen, “TurboADMM: A structure-exploiting parallel solver for multi-agent trajectory optimization,” in *Proceedings of the IEEE International Conference on Automation Science and Engineering*, 2026, best Paper Award Finalist.
- [2] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, “OSQP: An operator splitting solver for quadratic programs,” *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020.
- [3] H. J. Ferreau, C. Kirches, A. Potschka, H. G. Bock, and M. Diehl, “qpOASES: A parametric active-set algorithm for quadratic programming,” *Mathematical Programming Computation*, vol. 6, no. 4, pp. 327–363, 2014.
- [4] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, “Distributed optimization and statistical learning via the alternating direction method of multipliers,” *Foundations and Trends in Machine Learning*, vol. 3, no. 1, pp. 1–122, 2011.
- [5] C. E. Luis, M. Vukosavljev, and A. P. Schoellig, “Online trajectory generation with distributed model predictive control for multi-robot motion planning,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 604–611, 2020.
- [6] A. D. Saravanos, Y. Aoyama, H. Zhu, and E. A. Theodorou, “Distributed differential dynamic programming architectures for large-scale multiagent control,” *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4387–4407, 2023.
- [7] J. Tordesillas and J. P. How, “MADER: Trajectory planner in multiagent and dynamic environments,” *IEEE Transactions on Robotics*, vol. 38, no. 1, pp. 463–476, 2022.
- [8] K. Kondo, J. Tordesillas, R. Figueroa, J. Rached, J. Merkel, P. C. Lusk, and J. P. How, “Robust MADER: Decentralized and asynchronous multiagent trajectory planner robust to communication delay,” in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2023, pp. 1687–1693.
- [9] X. Zhou, J. Zhu, H. Zhou, C. Xu, and F. Gao, “EGO-Swarm: A fully autonomous and decentralized quadrotor swarm system in cluttered environments,” in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2021, pp. 4101–4107.
- [10] S. Wu, G. Chen, M. Shi, and J. Alonso-Mora, “Decentralized multi-agent trajectory planning in dynamic environments with spatiotemporal occupancy grid maps,” in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2024, pp. 7208–7214.
- [11] Y. Ma, Q. Khan, and D. Cremers, “MA-DV²F: A multi-agent navigation framework using dynamic velocity vector field,” *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 5823–5830, 2025.
- [12] J. Schulman, J. Ho, A. Lee, I. Awwal, H. Bradlow, and P. Abbeel, “Finding locally optimal, collision-free trajectories with sequential convex optimization,” in *Robotics: Science and Systems*, 2013.
- [13] Y. Mao, M. Szmuk, and B. Acikmese, “Successive convexification of non-convex optimal control problems and its convergence properties,” in *Proceedings of the IEEE Conference on Decision and Control*, 2016, pp. 3636–3641.
- [14] W. Hönig, J. A. Preiss, T. K. S. Kumar, G. S. Sukhatme, and N. Ayanian, “Trajectory planning for quadrotor swarms,” *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 856–869, 2018.
- [15] Y. Yang, S. Xu, X. Yan, J. Jiang, J. Wang, and H. Huang, “CSDO: Enhancing efficiency and success in large-scale multi-vehicle trajectory planning,” *IEEE Robotics and Automation Letters*, pp. 1–8, 2024.